

Motivation

One approach to clustering data is based on bipartite graph models, with nodes divided into **centers** and **members**. The edges to each center node specify its cluster memberships. Existing methods for this purpose have two key **limitations**:

- They require extra data about "center nodes" that **isn't always available**
- They often struggle with **heavy-tailed data** common in finance.

This paper proposes a new bipartite graph clustering method to overcome these challenges.

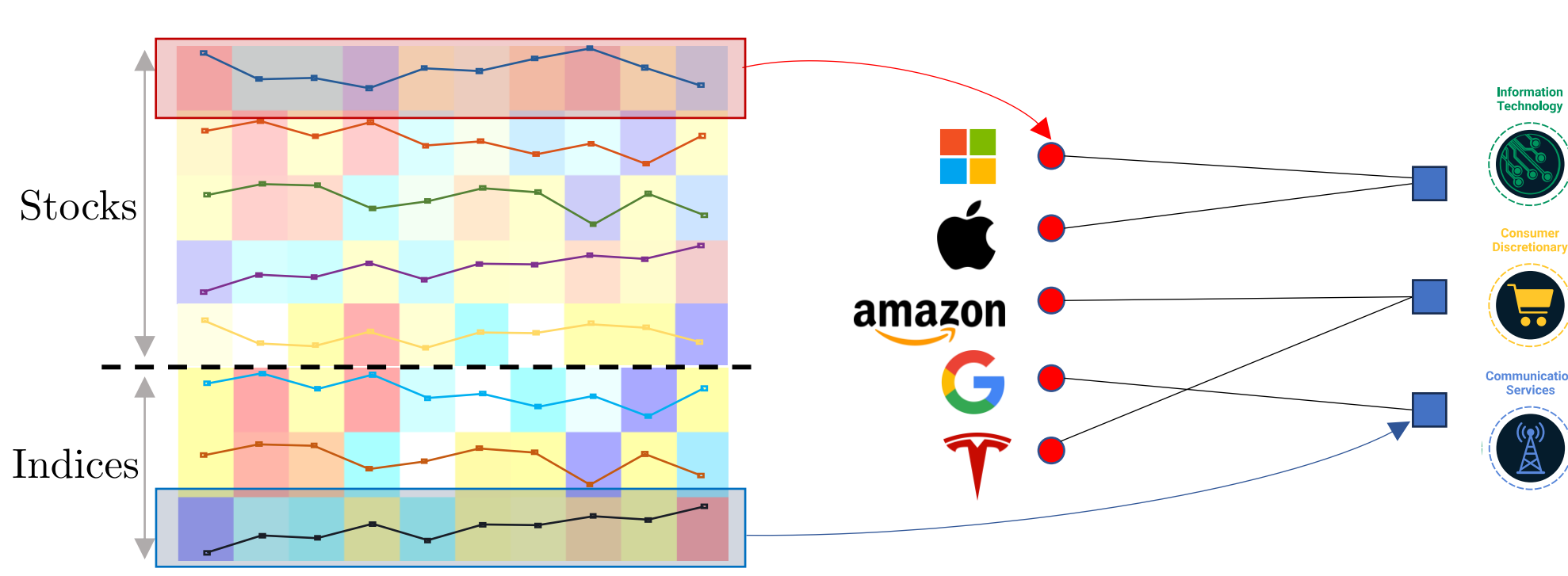


Figure 1. An example of bipartite graph clustering

Model

Graph Structure

- Consider a bipartite graph with undirected edges between centers and members with **weights given by $\mathbf{B} \in \mathbb{R}^{r \times k}$** .
- The Laplacian matrix $\mathbf{L}$ can be stated as
$$\mathbf{L} = \begin{bmatrix} \text{Diag}(\mathbf{B}\mathbf{1}_k) & -\mathbf{B} \\ -\mathbf{B}^\top & \text{Diag}(\mathbf{B}^\top\mathbf{1}_r) \end{bmatrix},$$
- Each $B_{i,j} \geq 0$ encodes the **probability** of member node i being within cluster j . So, $\mathbf{B}\mathbf{1}_k = \mathbf{1}_r$.
- We enforce **k disjoint clusters** or graph components (hard clustering). So, $\text{rank}(\mathbf{L}) = p - k$.

Signal Representation

- Values at nodes are represented by a **signal $\mathbf{x}_i \in \mathbb{R}^p$** (where $p = k + r$).
- We are given n measurements of the signal as **columns** of $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{p \times n}$.
- Assume $\mathbf{x}_i$ s are drawn from a multivariate **Student's t -distribution**:

$$p(\mathbf{x}_i) \propto \det^*(\mathbf{L})^{1/2} \left(1 + \frac{\mathbf{x}_i^\top \mathbf{L} \mathbf{x}_i}{\nu} \right)^{-(\nu+p)/2}, \quad \nu > 2.$$

- We aim to **maximize the log-likelihood** $\log p(\mathbf{X}|\mathbf{L})$
- This leads to a constrained optimization problem:

$$\min_{\mathbf{L}, \mathbf{B}} \frac{p+\nu}{n} \sum_{i=1}^n \log \left(1 + \frac{\mathbf{x}_i^\top \mathbf{L} \mathbf{x}_i}{\nu} \right) - \log \det^*(\mathbf{L})$$

$$\text{s. t. } (\mathbf{L}, \mathbf{B}) \in \Omega_C$$

- The **constraints Ω_C** are

$$\mathbf{L} = \begin{bmatrix} \text{Diag}(\mathbf{B}\mathbf{1}_k) & -\mathbf{B} \\ -\mathbf{B}^\top & \text{Diag}(\mathbf{B}^\top\mathbf{1}_r) \end{bmatrix}, \quad \text{rank}(\mathbf{L}) = p - k,$$

$$\mathbf{B} \geq 0, \quad \mathbf{B}\mathbf{1}_k = \mathbf{1}_r.$$

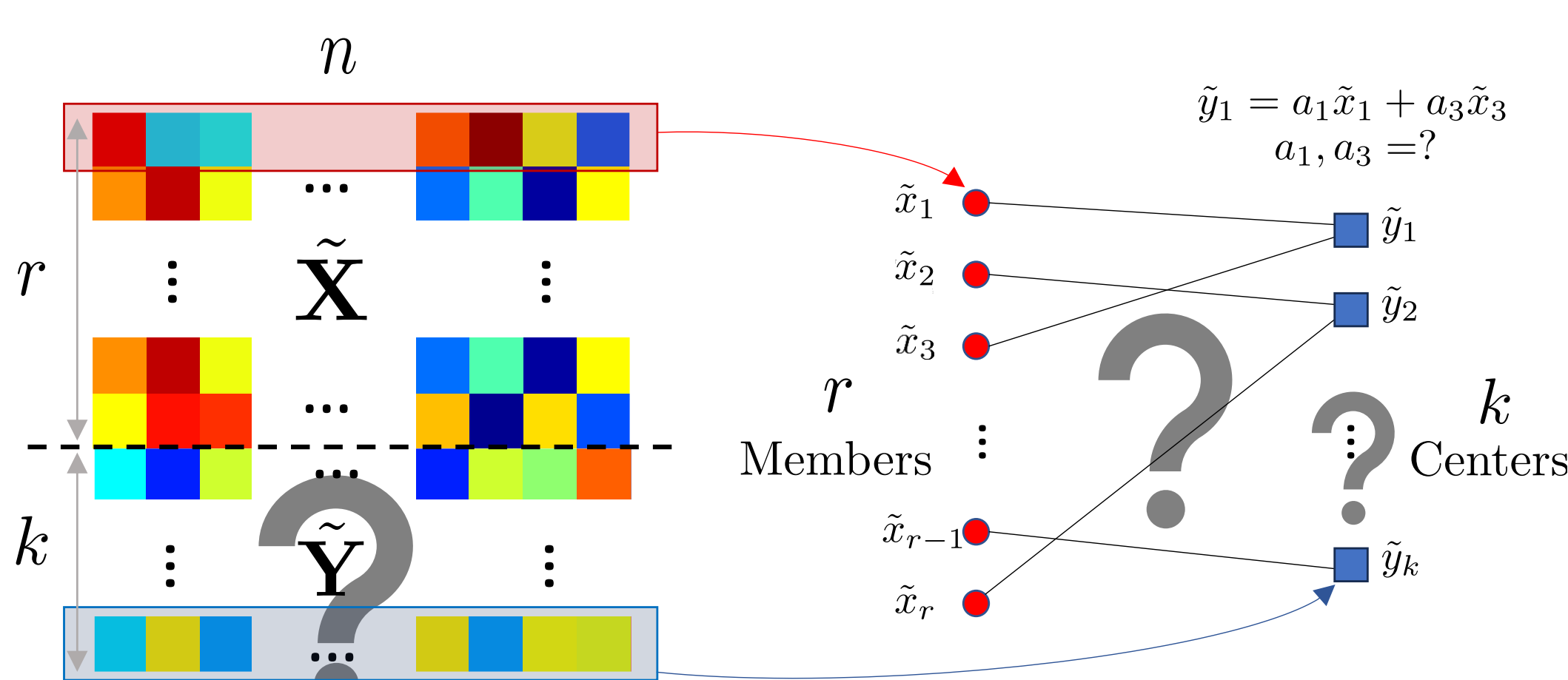


Figure 2. Bipartite graph clustering with incomplete data

Problem Statement

Handling Incomplete Data

- We only have **access to the first r rows** of $\mathbf{X}$ (members) denoted by $\tilde{\mathbf{X}} \in \mathbb{R}^{r \times n}$ and we need to **impute** $\tilde{\mathbf{Y}} \in \mathbb{R}^{k \times n}$ (center nodes data).
- Consider each center node data (row of $\tilde{\mathbf{Y}}$) is a **weighted average of its members** (rows of $\tilde{\mathbf{X}}$).
- This average is given by $\tilde{\mathbf{Y}} = \mathbf{A}^\top \tilde{\mathbf{X}}$ with $\mathbf{A} \in \mathbb{R}^{r \times k}$.

Problem Reformulation

- Restate $\mathbf{x}_i^\top = [\tilde{\mathbf{x}}_i^\top \tilde{\mathbf{x}}_i^\top \mathbf{A}^\top]^\top$, where $\tilde{\mathbf{x}}_i$ is the i -th column of $\tilde{\mathbf{X}}$
- Reformulate $\mathbf{x}_i^\top \mathbf{L} \mathbf{x}_i = h_i + \text{tr}(\mathbf{B} \mathbf{G}_i(\mathbf{A}))$ where

$$h_i = \text{tr}(\tilde{\mathbf{S}}_i), \quad \tilde{\mathbf{S}}_i = \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^\top,$$

$$\mathbf{G}_i(\mathbf{A}) = -2\mathbf{A}^\top \tilde{\mathbf{S}}_i + \text{diag}(\mathbf{A}^\top \tilde{\mathbf{S}}_i \mathbf{A}) \mathbf{1}_r^\top.$$

- Add $\mathbf{A}$ to the variables
- Add to the **constraints**: $\mathbf{A}^\top \mathbf{1}_r = \mathbf{1}_k$, $\mathbf{A} \geq 0$ (non-negative normalized weights)
- $\mathbf{A} \odot \mathbf{1}(\mathbf{B} = 0) = 0$ (same support with $\mathbf{B}$)
- The final problem would be

$$\min_{\mathbf{L}, \mathbf{B}, \mathbf{A}} \frac{p+\nu}{n} \sum_{i=1}^n \log \left(1 + \frac{h_i + \text{tr}(\mathbf{B} \mathbf{G}_i(\mathbf{A}))}{\nu} \right) - \log \det^*(\mathbf{L})$$

$$\text{s. t. } \mathbf{L} = \begin{bmatrix} \mathbf{I}_r & -\mathbf{B} \\ -\mathbf{B}^\top & \text{Diag}(\mathbf{B}^\top \mathbf{1}_r) \end{bmatrix}, \quad \text{rank}(\mathbf{L}) = p - k,$$

$$\mathbf{B}\mathbf{1}_k = \mathbf{1}_r, \quad \mathbf{B} \geq 0,$$

$$\mathbf{A}^\top \mathbf{1}_r = \mathbf{1}_k, \quad \mathbf{A} \geq 0, \quad \mathbf{A} \odot \mathbf{1}(\mathbf{B} = 0) = 0.$$

Proposed Algorithm

We solve this method using **in-exact ADMM**.

- We construct the augmented Lagrangian by **relaxing the constraint $\mathbf{L} = \begin{bmatrix} \mathbf{I}_r & -\mathbf{B} \\ -\mathbf{B}^\top & \text{Diag}(\mathbf{B}^\top \mathbf{1}_r) \end{bmatrix}$**

$$L_\rho(\mathbf{A}, \mathbf{B}, \mathbf{L}, \Phi) = \frac{p+\nu}{n} \sum_{i=1}^n \log \left(1 + \frac{h_i + \text{tr}(\mathbf{B} \mathbf{G}_i(\mathbf{A}))}{\nu} \right) - \log \det^*(\mathbf{L}) + \frac{\rho}{2} \left\| \mathbf{L} - \begin{bmatrix} \mathbf{I}_r & -\mathbf{B} \\ -\mathbf{B}^\top & \text{Diag}(\mathbf{B}^\top \mathbf{1}_r) \end{bmatrix} \right\|_F^2 + \langle \mathbf{L} - \begin{bmatrix} \mathbf{I}_r & -\mathbf{B} \\ -\mathbf{B}^\top & \text{Diag}(\mathbf{B}^\top \mathbf{1}_r) \end{bmatrix}, \Phi \rangle$$

$$\Phi$$
 is the **dual variable** (Lagrange multiplier)
 ρ is the ADMM **penalty parameter**.
- In each iteration, we consider 3 minimization subproblems, over $\mathbf{L}$, $\mathbf{B}$, and $\mathbf{A}$
- The solutions to the subproblems yield by
 - Majorization minimization (MM)** and
 - Projected gradient descent (PGD)****if no closed-form solution exists.**

- Input:** $\tilde{\mathbf{X}} \in \mathbb{R}^{r \times n}$ **Output:** $\mathbf{B}^l$
- Initialization:** $\mathbf{A}^0, \mathbf{B}^0 = \mathcal{P}_{\Omega_B}(-[(\mathbf{X}^\top \mathbf{X}/n)^\dagger]_{1:r, (r+1):p})$, $\Phi^0 = \mathbf{0}$, $l = 0$

repeat

- Update $\mathbf{L}^{l+1}$ with **closed-form solution**

$$\mathbf{L}^{l+1} = \underset{\mathbf{L} \geq 0, \text{rank}(\mathbf{L}) = p-k}{\text{argmin}} \frac{\rho}{2} \left\| \mathbf{L} - \begin{bmatrix} \mathbf{I}_r & -\mathbf{B}^l \\ -\mathbf{B}^{l\top} & \text{Diag}(\mathbf{B}^{l\top} \mathbf{1}_r) \end{bmatrix} \right\|_F^2 + \frac{1}{\rho} \Phi^l - \log \det^*(\mathbf{L})$$

- Update $\mathbf{B}^{l+1}$ by **iteratively solving**

$$\mathbf{B}^{l+1} = \underset{\mathbf{B} \geq 0, \mathbf{B}\mathbf{1}_k = \mathbf{1}_r}{\text{argmin}} \frac{p+\nu}{n} \sum_{i=1}^n \log \left(1 + \frac{h_i + \text{tr}(\mathbf{B} \mathbf{G}_i(\mathbf{A}^l))}{\nu} \right) + \frac{\rho}{2} \left\| \mathbf{L}^{l+1} - \begin{bmatrix} \mathbf{I}_r & -\mathbf{B} \\ -\mathbf{B}^\top & \text{Diag}(\mathbf{B}^\top \mathbf{1}_r) \end{bmatrix} \right\|_F^2 + \frac{1}{\rho} \Phi^l$$

- Update $\mathbf{A}^{l+1}$ by **iteratively solving**

$$\mathbf{A}^{l+1} = \underset{\mathbf{A} \geq 0, \mathbf{A}^\top \mathbf{1}_r = \mathbf{1}_k}{\text{argmin}} \frac{p+\nu}{n} \sum_{i=1}^n \log \left(1 + \frac{h_i + \text{tr}(\mathbf{B}^{l+1} \mathbf{G}_i(\mathbf{A}))}{\nu} \right)$$

$$\text{s. t. } \mathbf{A} \geq 0, \quad \mathbf{A}^\top \mathbf{1}_r = \mathbf{1}_k,$$

$$\mathbf{A} \odot \mathbf{1}(\mathbf{B}^{l+1} = 0) = 0$$

- Update the dual variable:

$$\Phi^{l+1} = \Phi^l + \rho \left(\mathbf{L}^{l+1} - \begin{bmatrix} \mathbf{I}_r & -\mathbf{B}^{l+1} \\ -\mathbf{B}^{l+1\top} & \text{Diag}(\mathbf{B}^{l+1\top} \mathbf{1}_r) \end{bmatrix} \right)$$

- Set $l \leftarrow l + 1$.

- until** a stopping criterion is satisfied

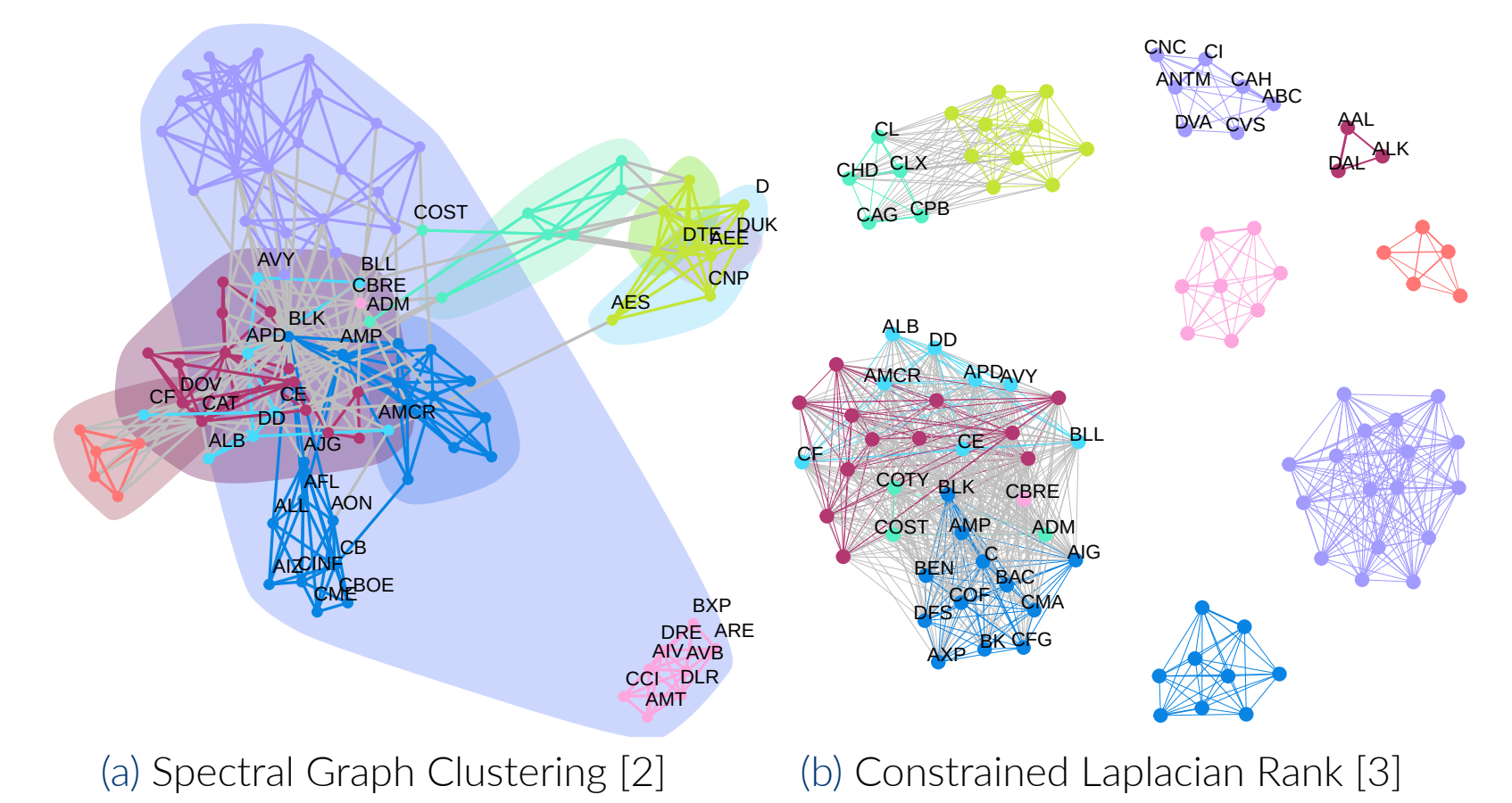
Numerical Results

We test on real S&P500 stocks log-return data. We evaluate the **clustering performance** (with true labels given by GICS standard [1]) in terms of accuracy (ACC), modularity (MOD), purity, adjusted rand index (ARI) and the Calinski-Harabasz index (CHI).

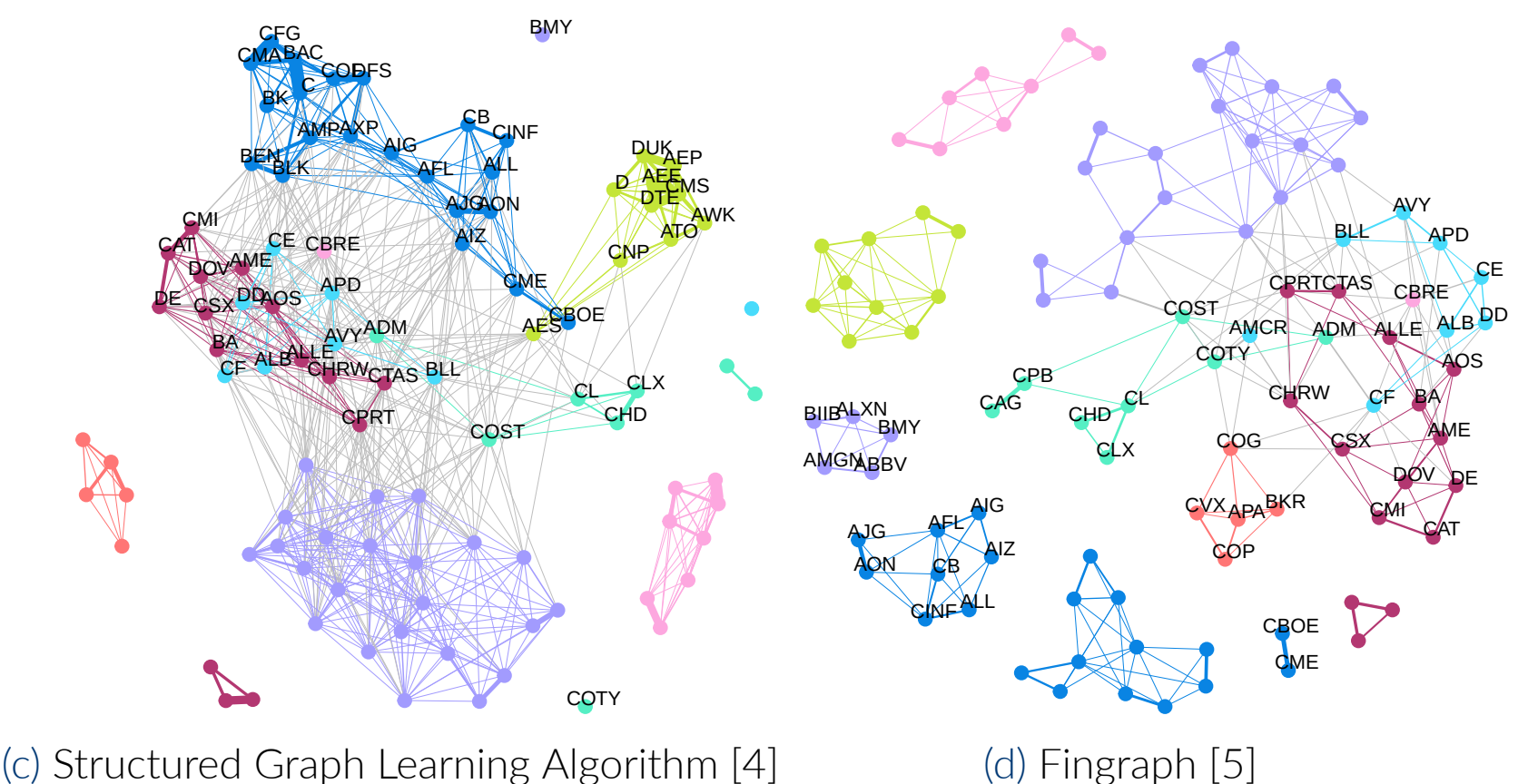
We try **different initializations for $\mathbf{A}$** : one with uniform and another with Gaussian normal distributions.

Table 1. Clustering performance of the S&P500 stock return dataset

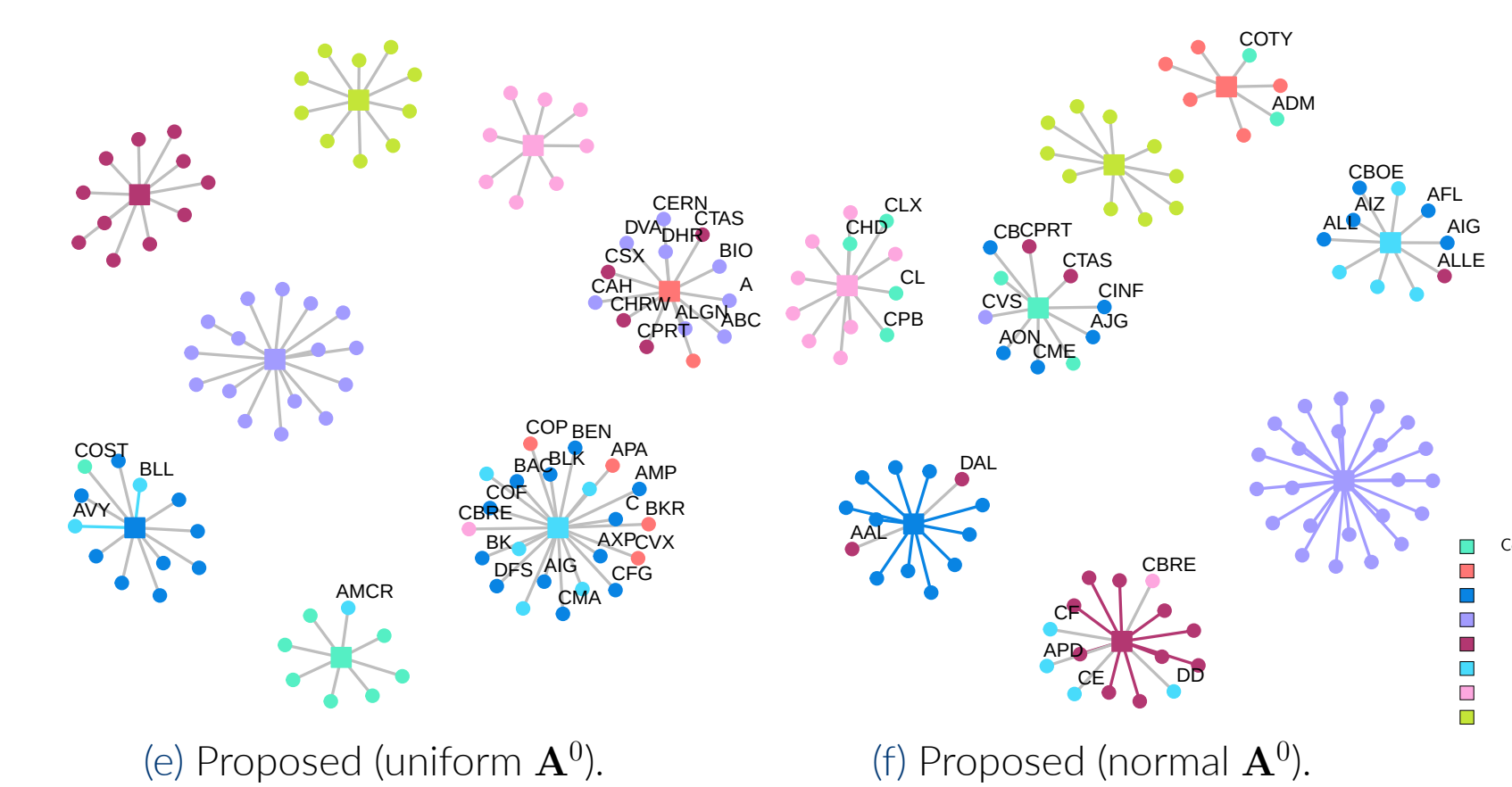
	ACC $\uparrow$	Purity $\uparrow$	MOD $\uparrow$	ARI $\uparrow$	CHI $\uparrow$
SGC [2]	0.62	0.68	0.52	0.42	3.95
CLR [3]	0.61	0.71	0.34	0.43	4.73
SGLA [4]	0.42	0.44	0.41	0.12	3.08
Fingraph [5]	0.51	0.66	0.64	0.26	3.48
Proposed (uniform $\mathbf{A}^0$)	0.67	0.81	0.85	0.53	5.05
Proposed (normal $\mathbf{A}^0$)	0.73	0.77	0.85	0.63	5.05



(a) Spectral Graph Clustering [2] (b) Constrained Laplacian Rank [3]

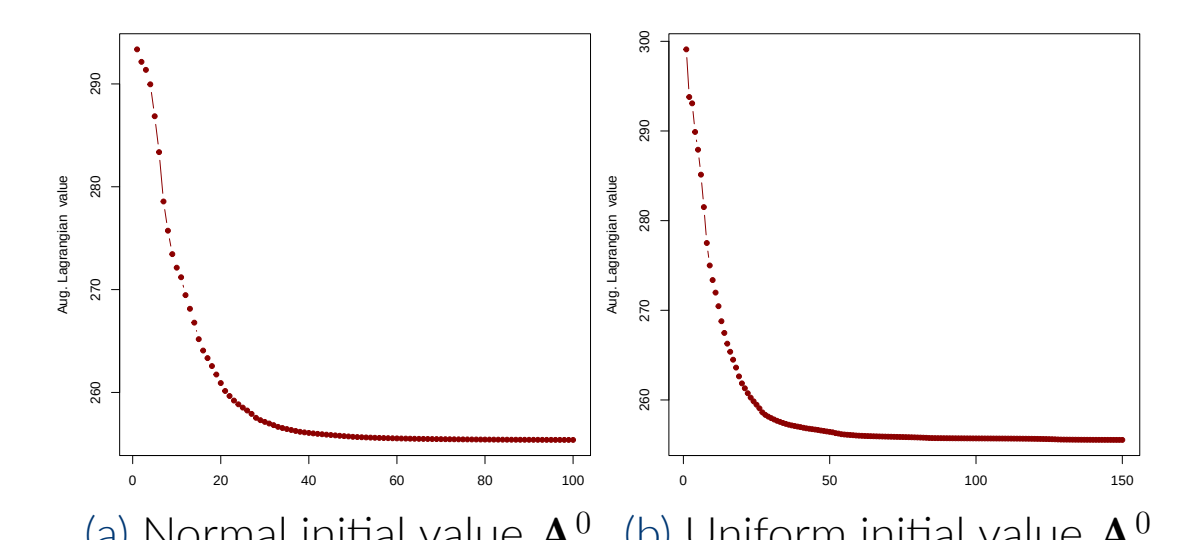


(c) Structured Graph Learning Algorithm [4] (d) Fingraph [5]



(e) Proposed (uniform $\mathbf{A}^0$). (f) Proposed (normal $\mathbf{A}^0$).

Figure 3. The graphs learned from log-returns data of 100 stocks in S&P500 index (including $k = 8$ sectors).



(a) Normal initial value $\mathbf{A}^0$ (b) Uniform initial value $\mathbf{A}^0$

Figure 4. The convergence of the proposed method for different values of $\mathbf{A}^0$.

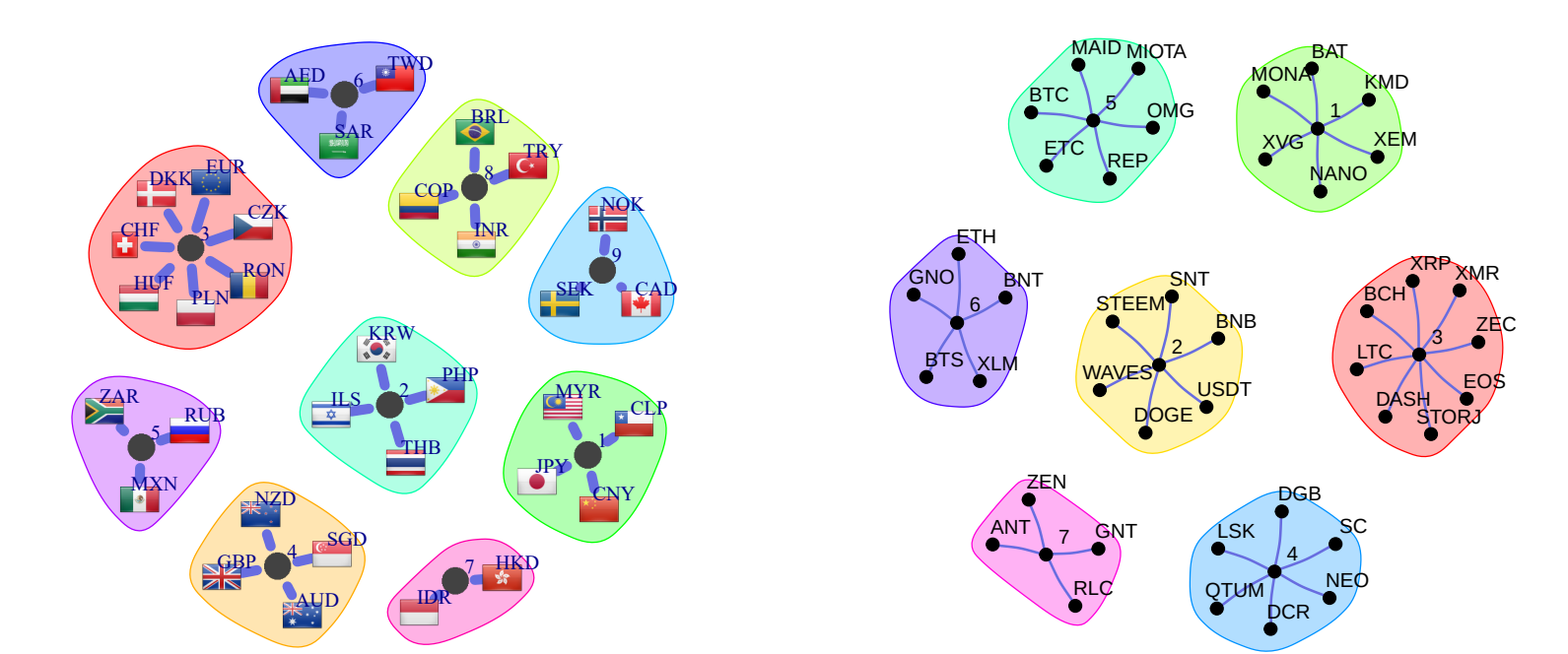


Figure 5. More results of the proposed method on forex and crypto data

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